

Oracle Database 19c ASM Logical Backup Guide

Comprehensive Step-by-Step Documentation for Full Database, Tablespace, Schema, and Table Level Exports

1. Overview & Prerequisites

When working with Oracle databases utilizing Automatic Storage Management (ASM) for physical file storage, logical backups are performed using Oracle Data Pump (expdp). Data Pump writes export dump files to standard operating system directories rather than direct ASM disk groups.

Environment Details: Oracle Database 19c Enterprise Edition on Linux (ASM storage instance), running as user oracle with asminstall/asmadmin privileges.

2. Full Database Backup in ASM Instance

To take a complete logical backup of the entire Oracle database residing on ASM storage, follow the structured OS and SQL steps below.

Step 1: Create Backup Directory at OS Level

Log in as root or sudo user to create the target backup directory on the filesystem and assign correct ownership and permissions.

```
[root@localhost oracle]# mkdir -p /u02/lbkp
[root@localhost oracle]# chown -R oracle:oinstall /u02/lbkp
[root@localhost oracle]# chmod -R 775 /u02/lbkp
```

Step 2: Create a Logical Directory in Oracle

Connect to SQL*Plus as SYSDBA and map the OS directory to a database directory object.

```
SQL> create or replace directory exp_dir as '/u02/lbkp';
Directory created.
```

Step 3: Grant Read/Write Permissions

Grant necessary privileges on the directory object to the backup operator or SYSTEM user.

```
SQL> grant read,write on directory exp_dir to system;
Grant succeeded.
```

Step 4: Execute Full Database Export

Run the expdp utility from the OS command line with FULL=y and parallel worker processes enabled for high performance.

```
[oracle@localhost lbkp]$ expdp system/oracle directory=exp_dir dumpfile=full.dmp
full=y parallel=4
```

```
Export: Release 19.0.0.0.0 - Production on Mon Sep 7 12:49:29 2026
Version 19.3.0.0.0
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Starting "SYSTEM"."SYS_EXPORT_FULL_01": system/***** directory=exp_dir
dumpfile=full.dmp full=y parallel=4
```

Step 5: Verify Backup Completion

Check the export directory to ensure the dump file and log file have been successfully generated with expected sizes.

```
[oracle@localhost lbkp]$ ls -lh
```

```
total 4.0M
-rw-r--r--. 1 oracle asmadmin 11K Sep  7 13:00 export.log
-rw-r-----. 1 oracle asmadmin 4.0M Sep  7 13:00 full.dmp
```

3. Schema-Level Backup

To back up a specific user schema (e.g., hr) instead of the entire database, use the SCHEMAS parameter.

Execute Schema Export

```
[oracle@localhost lbkp]$ expdp system/oracle directory=exp_dir dumpfile=hr.dmp  
schemas=hr
```

```
Export: Release 19.0.0.0.0 - Production on Mon Sep 7 13:06:23 2026  
Version 19.3.0.0.0  
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.  
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Starting "SYSTEM"."SYS_EXPORT_SCHEMA_01": system/***** directory=exp_dir  
dumpfile=hr.dmp schemas=hr  
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
```

Verify Schema Backup

```
[oracle@localhost lbkp]$ ls -lh
```

```
total 4.7M  
-rw-r--r--. 1 oracle asmadmin 2.4K Sep  7 13:08 export.log  
-rw-r-----. 1 oracle asmadmin 4.0M Sep  7 13:00 full.dmp  
-rw-r-----. 1 oracle asmadmin 708K Sep  7 13:08 hr.dmp
```

4. Tablespace-Level Backup

To back up a specific tablespace (e.g., users) using parallel workers and wildcard dump file naming (`%U`), use the TABLESPACES parameter.

Execute Tablespace Export

```
[oracle@localhost lbkp]$ expdp system/oracle directory=exp_dir  
dumpfile=user_%U.dmp tablespaces=users parallel=2
```

```
Export: Release 19.0.0.0.0 - Production on Mon Sep 7 13:23:46 2026  
Version 19.3.0.0.0  
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.  
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
```

Verify Tablespace Backup

```
[oracle@localhost lbkp]$ ls -lh
```

```
total 4.9M
-rw-r--r--. 1 oracle asmadmin 1.2K Sep  7 13:30 export.log
-rw-r-----. 1 oracle asmadmin 4.0M Sep  7 13:00 full.dmp
-rw-r-----. 1 oracle asmadmin 708K Sep  7 13:08 hr.dmp
-rw-r-----. 1 oracle asmadmin 212K Sep  7 13:30 user_01.dmp
```

5. Table-Level Backup

To back up an individual table (e.g., `scott.employees`), use the `TABLES` parameter.

Execute Table Export

```
[oracle@localhost lbkp]$ expdp system/oracle directory=exp_dir
dumpfile=employees_%U.dmp tables=scott.employees
```

```
Export: Release 19.0.0.0.0 - Production on Mon Sep 7 13:40:04 2026
Version 19.3.0.0.0
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.
Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Starting "SYSTEM"."SYS_EXPORT_TABLE_01": system/***** directory=exp_dir
dumpfile=employees_%U.dmp tables=scott.employees
Processing object type TABLE_EXPORT/TABLE/TABLE_DATA
```

Verify Final Backup Directory Contents

```
[oracle@localhost lbkp]$ ls -lh
```

```
total 5.1M
-rw-r-----. 1 oracle asmadmin 212K Sep  7 13:40 employees_01.dmp
-rw-r--r--. 1 oracle asmadmin 1.2K Sep  7 13:40 export.log
-rw-r-----. 1 oracle asmadmin 4.0M Sep  7 13:00 full.dmp
-rw-r-----. 1 oracle asmadmin 708K Sep  7 13:08 hr.dmp
-rw-r-----. 1 oracle asmadmin 212K Sep  7 13:30 user_01.dmp
```